

Implement Life Cycle Policies

1. Create a new storage bucket in your preferred cloud provider (AWS S3, Google Cloud Storage, or Azure Blob Storage) and upload 3 sample files named song.mp3, food.jpg, and bill.pdf, simulating files from apps like Spotify, Zomato, and Paytm.

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1. Log in to the **AWS Management Console**.
2. Open **Amazon S3**.
3. Click **Create Bucket**.
4. Enter a unique bucket name (for example, my-cloud-storage-deep-20260718).
5. Select the **Asia Pacific (Mumbai)** region.
6. Click **Create Bucket**.
7. Open the bucket and upload the following files:\

File Name Represents

song.mp3 Spotify music file

food.jpg Zomato food image

bill.pdf Paytm payment receipt

➤ **Result:** All three files are successfully uploaded to the S3 bucket.

2. Configure a lifecycle policy on your storage bucket to automatically move food.jpg to a cheaper storage class (like S3 Glacier, Google Coldline, or Azure Cool) after 1 day and delete bill.pdf after 3 days.

Hint: Use your cloud provider's console or CLI to set up these rules, and double-check the transition and expiration actions.

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1. Open the bucket and go to the Management tab.
2. Click Create Lifecycle Rule.
3. Rule Name: MoveFoodImageToGlacier
4. Apply the rule only to the object:
 - food.jpg
5. Configure the transition:

- Transition to: S3 Glacier Flexible Retrieval
- After: 1 day

Rule 2: Delete bill.pdf after 3 days

1. Create another Lifecycle Rule.
2. Rule Name: DeleteBillPDF
3. Apply the rule only to the object:

bill.pdf

4. Configure the expiration:
 - Delete object after: 3 days

Result:

- food.jpg moves to Glacier after 1 day.
 - bill.pdf is automatically deleted after 3 days.
4. Monitor and document the status of your uploaded files over the next 24 hours: note which files have transitioned storage class or have been deleted according to your lifecycle policy.

Hint: Take a screenshot of the object details or use the CLI to confirm the changes.

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- After waiting for the configured time period:

File	Expected Status
song.mp3	Remains in S3 Standard
food.jpg	Storage class changes to Glacier
bill.pdf	Deleted automatically

- **Verification**

- Check the object details in the S3 Console or use the AWS CLI:
 - `aws s3 ls s3://your-bucket-name`
 - Document the observed changes and include screenshots of the object details or CLI output.
5. Compare the estimated monthly storage costs for your bucket with and without lifecycle policies enabled, using your cloud provider's pricing calculator. Write a short note on

how lifecycle policies can help apps like Instagram or BookMyShow save money at scale.

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Example Monthly Storage Cost (5 GB)

Configuration	Estimated Monthly Cost (INR)*
S3 Standard Only	₹10–₹15
With Lifecycle Policy (Glacier)	₹2–₹5

- Actual prices depend on the AWS region and current pricing.
6. Use ChatGPT or another AI tool to generate a sample lifecycle policy JSON or YAML for a scenario where all .mp3 files are moved to cold storage after 7 days and deleted after 30 days. Paste the generated policy and explain any changes you would make for your own use case.

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❖ Explanation

Section	Description
ID	The unique name of the lifecycle rule.
Status	Enables the lifecycle rule.
Filter	Determines which objects the rule applies to. For a production environment, this could be changed to an .mp3 prefix or a dedicated folder.
Transitions	Moves matching objects to Glacier after 7 days.
Expiration	Permanently deletes the objects after 30 days.

- Instead of applying the rule to every object, I would store music files in a dedicated folder such as **music/** and apply the lifecycle rule only to that folder. This prevents images, PDFs, and other important files from being moved to Glacier or deleted unintentionally.